

Unsupervised Machine Learning for Ferroelectric Nanodots

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1 Introduction

Ferroelectric nanodots have emerged as a promising class of nanoscale materials due to their unique polarization properties, size-dependent phase transitions, and potential applications in high-density data storage, nanoelectronics, and energy-efficient devices [1]. At reduced dimensions, ferroelectric systems exhibit complex domain structures and emergent phenomena that are not present in their bulk counterparts[2]. Understanding these behaviors is critical for advancing next-generation technologies, yet it remains a significant challenge due to the intricate interplay of electrostatic, elastic, and thermal effects at the nanoscale.

Traditional approaches to studying ferroelectric nanodots rely heavily on theoretical modeling, numerical simulations, and experimental characterization techniques such as piezoresponse force microscopy[3]. While these methods have provided valuable insights, they often require prior assumptions about domain configurations or phase boundaries, limiting their ability to uncover hidden patterns in large and high-dimensional datasets. With the increasing availability of simulation and experimental data, there is a growing need for data-driven approaches that can autonomously identify underlying structures without explicit supervision[4].

Unsupervised machine learning has emerged as a powerful tool for extracting meaningful patterns from complex datasets without labeled outputs. Techniques such as clustering, dimensionality reduction, and manifold learning enable the identification of intrinsic structures, phase transitions, and correlations within data[5]. In the context of ferroelectric nanodots, these methods can be used to classify domain configurations, detect phase boundaries, and reveal previously unrecognized states of polarization.

This study explores the application of unsupervised machine learning techniques to analyze ferroelectric nanodot systems. By leveraging methods such as k-means clustering and principal component analysis (PCA), we aim to un-

cover latent patterns in polarization data and provide a deeper understanding of nanoscale ferroelectric behavior[6]. The integration of machine learning with physical modeling not only enhances interpretability but also opens new avenues for discovering emergent phenomena in complex material systems.

2 Result and Discussion

The application of unsupervised machine learning techniques to ferroelectric nanodot data reveals meaningful structural patterns and distinct polarization states. To determine the optimal number of clusters, the distortion and inertia metrics were evaluated as functions of the number of clusters K . Both curves exhibit a sharp decrease from $K=1$ to $K=3$, followed by a gradual reduction for higher values of K . This characteristic “elbow” behavior suggests that the intrinsic structure of the dataset is best captured within the range of $K=3$ to $K=4$ (See Figure 1).

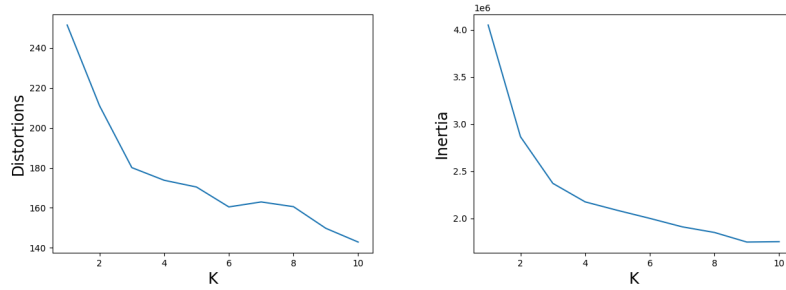


Figure 1: Elbow plots for determining optimal number of clusters.

To further interpret the clustering results, Principal Component Analysis (PCA) was employed to project the high-dimensional polarization data into a two-dimensional space defined by the first two principal components (PC1 and PC2). This transformation preserves the dominant variance in the dataset and enables clear visualization of the cluster distributions. For $K=3$ (Figure 2), the data points form three well-separated clusters in PCA space, indicating the presence of three dominant polarization configurations in the ferroelectric nanodots. The clear spatial separation suggests strong inherent grouping and validates the effectiveness of the clustering approach.

When the number of clusters is increased to $K=4$, one of the previously identified clusters splits into two distinct subgroups. This indicates the presence of finer structural variations within a broader polarization state, revealing additional complexity in the system. The clusters remain relatively well-separated, suggesting that $K=4$ still provides physically meaningful insights. However, increasing the number of clusters further to $K=5$ results in fragmentation of the data into smaller groups, some of which are sparsely populated or less distinct. This behavior suggests that the clustering algorithm begins to capture

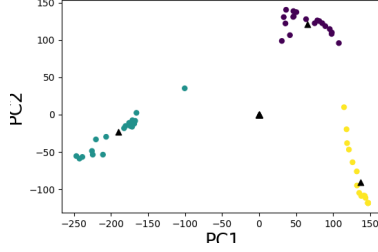


Figure 2: PC1-PC2 Analysis for K=3

noise or minor fluctuations rather than significant physical differences, reducing interpretability(See Figure 3).

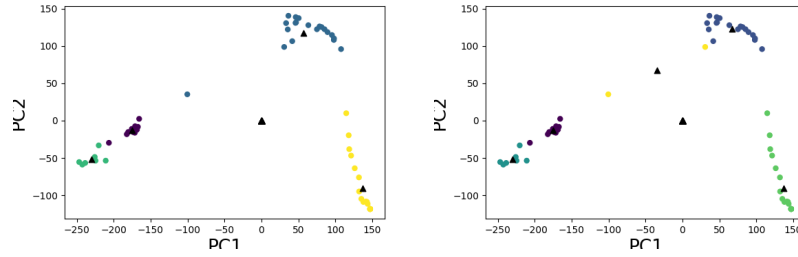


Figure 3: PC1-PC2 Analysis for K=4 and K=5.

The variation of cluster labels with temperature reveals thermally induced transitions between distinct polarization states, demonstrating that unsupervised learning effectively captures phase behavior in ferroelectric nanodots(See Figure 4).

3 Conclusion

This study demonstrates that unsupervised machine learning, particularly K-means clustering combined with Principal Component Analysis, is highly effective in analyzing ferroelectric nanodot systems. The results reveal that a small number of clusters (K=3-4) captures the essential polarization states, while temperature-dependent clustering highlights underlying phase transitions and domain evolution. Overall, the approach provides a powerful data-driven framework for uncovering hidden patterns and enhancing the understanding of nanoscale ferroelectric behavior.

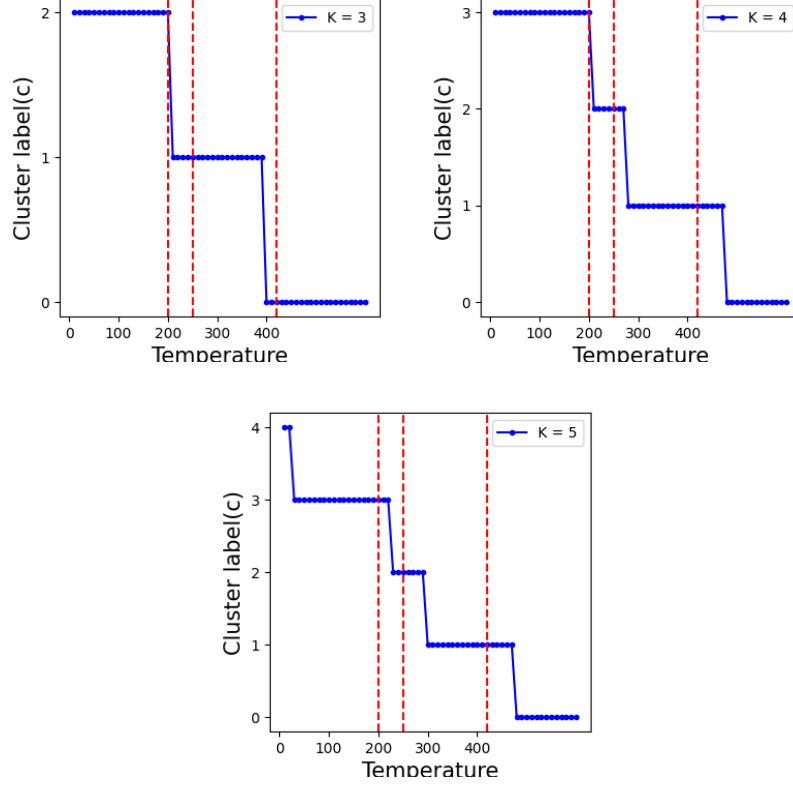


Figure 4: Temperature Cluster Label relation

4 References

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